NAME:Abhishek Yadav.ML LAB-6 Aim: To design and implement an Artificial Neural Network for classification problem and evaluate its learning performance.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
import pandas as pd
path="/content/drive/MyDrive/ml_dataset6/churn.csv"
df=pd.read_csv(path)
df.head(5)
```

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMemb
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	
1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	
2	3	15619304	Onio	502	France	Female	42	8	159660.80	3	1	
3	4	15701354	Boni	699	France	Female	39	1	0.00	2	0	
4	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	

```
df.shape
```

```
(10000, 14)
```

```
df.columns
```

```
Index(['RowNumber', 'CustomerId', 'Surname', 'CreditScore', 'Geography',
       'Gender', 'Age', 'Tenure', 'Balance', 'NumOfProducts', 'HasCrCard',
       'IsActiveMember', 'EstimatedSalary', 'Exited'],
      dtype='object')
```

```
df.isnull().any().any()
```

```
np.False_
```

```
df.isnull().sum()
```

	0
RowNumber	0
CustomerId	0
Surname	0
CreditScore	0
Geography	0
Gender	0
Age	0
Tenure	0
Balance	0
NumOfProducts	0
HasCrCard	0
IsActiveMember	0
EstimatedSalary	0
Exited	0

dtype: int64

```
df[df.isnull().any(axis = 1)]
```

RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember

```
x = df.drop(columns=['Exited'])
y = df['Exited']
```

```
x.head()
```

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMemb
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	
1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	
2	3	15619304	Onio	502	France	Female	42	8	159660.80	3	1	
3	4	15701354	Boni	699	France	Female	39	1	0.00	2	0	
4	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	

```
x = pd.get_dummies(x, dtype = int)
```

```
x.head()
```

	RowNumber	CustomerId	CreditScore	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	...	Su
0	1	15634602	619	42	2	0.00	1	1	1	101348.88	...	
1	2	15647311	608	41	1	83807.86	1	0	1	112542.58	...	
2	3	15619304	502	42	8	159660.80	3	1	0	113931.57	...	
3	4	15701354	699	39	1	0.00	2	0	0	93826.63	...	
4	5	15737888	850	43	2	125510.82	1	1	1	79084.10	...	

5 rows × 2947 columns

```
x.shape
```

```
(10000, 2947)
```

```
x.columns
```

```
Index(['RowNumber', 'CustomerId', 'CreditScore', 'Age', 'Tenure', 'Balance',
       'NumOfProducts', 'HasCrCard', 'IsActiveMember', 'EstimatedSalary',
       ...
       'Surname_Zubarev', 'Surname_Zubareva', 'Surname_Zuev', 'Surname_Zuyev',
       'Surname_Zuyeva', 'Geography_France', 'Geography_Germany',
       'Geography_Spain', 'Gender_Female', 'Gender_Male'],
      dtype='object', length=2947)
```

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x,y, train_size = 0.7, test_size=0.3, random_state = 42)
x_train.shape, x_test.shape
```

```
((7000, 2947), (3000, 2947))
```

```
from sklearn.ensemble import RandomForestClassifier
rf_classifier = RandomForestClassifier(n_estimators=100, n_jobs=-1,max_depth=5, oob_score=True, random_state=42)
rf_classifier.fit(x_train, y_train)
```

```
RandomForestClassifier
RandomForestClassifier(max_depth=5, n_jobs=-1, oob_score=True, random_state=42)
```

```
y_pred = rf_classifier.predict(x_test)
```

```
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
print('confusion matrix')
cm=confusion_matrix(y_test, y_pred)
print(confusion_matrix(y_test, y_pred))
```

```
confusion matrix
[[2416    0]
 [ 584    0]]
```

```
print('Accuracy Score:')
print(accuracy_score(y_test, y_pred))
```

```
Accuracy Score:
0.8053333333333333
```

```
print(rf_classifier.oob_score)
```

```
True
```

```
print('Classification Report:')
print(classification_report(y_test, y_pred))
```

```
Classification Report:
              precision    recall  f1-score   support

           0       0.81      1.00      0.89      2416
           1       0.00      0.00      0.00       584

    accuracy                           0.81      3000
   macro avg       0.40      0.50      0.45      3000
weighted avg       0.65      0.81      0.72      3000

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-define
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-define
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-define
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

```
x = df.iloc[:,3:-1]
x
```

	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary
0	619	France	Female	42	2	0.00	1	1	1	101348.88
1	608	Spain	Female	41	1	83807.86	1	0	1	112542.58
2	502	France	Female	42	8	159660.80	3	1	0	113931.57
3	699	France	Female	39	1	0.00	2	0	0	93826.63
4	850	Spain	Female	43	2	125510.82	1	1	1	79084.10
...	...	...	...	...	...	...	...	...	...	...
9995	771	France	Male	39	5	0.00	2	1	0	96270.64
9996	516	France	Male	35	10	57369.61	1	1	1	101699.77
9997	709	France	Female	36	7	0.00	1	0	1	42085.58
9998	772	Germany	Male	42	3	75075.31	2	1	0	92888.52
9999	792	France	Female	28	4	130142.79	1	1	0	38190.78

10000 rows × 10 columns

```
y=df.iloc[:,-1]
y
```

	Exited
0	1
1	0
2	1
3	0
4	0
...	...
9995	0
9996	0
9997	1
9998	1
9999	0

10000 rows × 1 columns

dtype: int64

```
x = pd.get_dummies(x, dtype = int)
x
```

	CreditScore	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Geography_France	Geography
0	619	42	2	0.00	1	1	1	101348.88	1	
1	608	41	1	83807.86	1	0	1	112542.58	0	
2	502	42	8	159660.80	3	1	0	113931.57	1	
3	699	39	1	0.00	2	0	0	93826.63	1	
4	850	43	2	125510.82	1	1	1	79084.10	0	
...	...	...	...	...	...	...	...	...	...	
9995	771	39	5	0.00	2	1	0	96270.64	1	
9996	516	35	10	57369.61	1	1	1	101699.77	1	
9997	709	36	7	0.00	1	0	1	42085.58	1	
9998	772	42	3	75075.31	2	1	0	92888.52	0	
9999	792	28	4	130142.79	1	1	0	38190.78	1	

10000 rows × 13 columns

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, train_size=0.7, test_size=0.3, random_state=42)
x_train.shape, x_test.shape
```

```
((7000, 13), (3000, 13))
```

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
x_train = sc.fit_transform(x_train)
x_test = sc.transform(x_test)
```

```
x_train
```

```
array([[-0.34459497, -0.65674999, -0.34217046, ...,  1.73073215,
         1.09823226, -1.09823226],
       [-0.09518109, -0.46637979,  0.69816249, ..., -0.57779016,
         1.09823226, -1.09823226],
       [-0.94734518, -0.56156489,  0.35138484, ..., -0.57779016,
        -0.91055421,  0.91055421],
       ...,
       [ 0.86090545, -0.08563939, -1.38250341, ..., -0.57779016,
         1.09823226, -1.09823226],
       [ 0.15423279,  0.39028611,  1.04494014, ..., -0.57779016,
        -0.91055421,  0.91055421],
       [ 0.46600014,  1.1517669 , -1.38250341, ..., -0.57779016,
        -0.91055421,  0.91055421]])
```

```
import numpy as np
import pandas as pd
import tensorflow as tf
```

```
ann = tf.keras.models.Sequential()
```

```
ann.add(tf.keras.layers.Dense(units=6, activation='relu'))
```

```
ann.add(tf.keras.layers.Dense(units=6, activation='relu'))
```

```
ann.add(tf.keras.layers.Dense(units=1, activation='sigmoid'))
```

```
ann.compile(optimizer = 'adam', loss = 'binary_crossentropy', metrics = ['accuracy'])
```

```
ann.fit(x_train, y_train, batch_size = 32, epochs = 100)
```

```
Epoch 1/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.7780 - loss: 0.5578
Epoch 2/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.7900 - loss: 0.4633
Epoch 3/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8020 - loss: 0.4335
Epoch 4/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8091 - loss: 0.4282
Epoch 5/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8265 - loss: 0.3974
Epoch 6/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.8373 - loss: 0.3856
Epoch 7/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8462 - loss: 0.3715
Epoch 8/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8472 - loss: 0.3624
Epoch 9/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8570 - loss: 0.3532
Epoch 10/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8466 - loss: 0.3612
Epoch 11/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8621 - loss: 0.3433
Epoch 12/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8555 - loss: 0.3476
Epoch 13/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.8562 - loss: 0.3509
Epoch 14/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8592 - loss: 0.3437
Epoch 15/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8566 - loss: 0.3454
Epoch 16/100
```

```
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8572 - loss: 0.3485
Epoch 17/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.8583 - loss: 0.3485
Epoch 18/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8526 - loss: 0.3579
Epoch 19/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8595 - loss: 0.3448
Epoch 20/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8658 - loss: 0.3329
Epoch 21/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8582 - loss: 0.3491
Epoch 22/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8565 - loss: 0.3461
Epoch 23/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8674 - loss: 0.3282
Epoch 24/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8561 - loss: 0.3485
Epoch 25/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8598 - loss: 0.3507
Epoch 26/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8573 - loss: 0.3470
Epoch 27/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8659 - loss: 0.3276
Epoch 28/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8583 - loss: 0.3427
Epoch 29/100
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8557 - loss: 0.3471
```

```
y_pred = ann.predict(x_test)
y_pred = (y_pred > 0.5)
y_pred
```

```
94/94 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step
array([[False],
       [False],
       [False],
       ...,
       [False],
       [False],
       [False]])
```

```
print('confusion matrix')
cm=confusion_matrix(y_test, y_pred)
print(confusion_matrix(y_test, y_pred))
```

```
confusion matrix
[[2317   99]
 [ 315  269]]
```

```
print('Accuracy Score:')
print(accuracy_score(y_test, y_pred))
```

```
Accuracy Score:
0.862
```

```
print('Classification Report')
print(classification_report(y_test, y_pred))
```

```
Classification Report
```